Problem Set Errors 1: Worked Solutions

Frame and packet error probability · Data Link Errors lecture, slides 6–8

Formulas used: P(frame error) $\mathbf{p = 1 - (1 - p_b)^L}$ · approximation $\mathbf{p \approx L \times p_b}$ (only when $p \ll 1$) · over k hops, replace L by $\mathbf{L \times k}$ · working backwards: $\mathbf{L < \ln(1 - p) \div \ln(1 - p_b)}$ · L is in **bits** (bytes × 8) · hops = intermediate nodes + 1

Q1

> Frame of 12,000 bits, probability of bit error 1 in a million. What is the probability of frame error?

1. L = 12,000 bits and $p_b = 10^{-6}$.
2. Approximation: $p \approx 12{,}000 \times 10^{-6} = 0.012$. That's small, so the approximation is valid.
3. Exact check: $1 - (1 - 10^{-6})^{12{,}000} = 0.01193$.

$p \approx 0.012$ (1.2%)

Q2

> Frame of 6 kB, probability of bit error 1 in a million. What is the probability of frame error?

1. Convert: 6 kB = 6,000 B × 8 = 48,000 bits.
2. Approximation: $p \approx 48{,}000 \times 10^{-6} = 0.048$. That's small, so it's valid.
3. Exact: $1 - (1 - 10^{-6})^{48{,}000} = 0.0469$.

$p \approx 0.048$ (about 4.7–4.8%)

> If 1 kB is taken as 1,024 B, then L = 49,152 bits and $p \approx 0.049$.

Q3

> Probability of bit error 1 in a billion, frame error below 1 in a million. What is the largest frame size in bytes?

1. Condition: $L \times p_b < p$, so $L < 10^{-6} \div 10^{-9} = 1{,}000$ bits.
2. The target is tiny, so the approximation is valid. The exact formula also gives 1,000 bits.
3. Convert: 1,000 ÷ 8 = 125 B.

125 bytes

Q4

> Probability of bit error 1 in a million, frame error below 1 in 10. What is the largest frame size in bytes?

1. Approximation: $L < 0.1 \div 10^{-6} = 100{,}000$ bits = 12,500 B.
2. But 0.1 isn't $\ll 1$, so use the exact formula: $L < \ln(0.9) \div \ln(1 - 10^{-6}) = 105{,}360$ bits.
3. Convert: 105,360 ÷ 8 = 13,170 B, rounded down to stay under the target.

≈ 13,170 bytes (exact)

> The approximation gives 12,500 B, about 5% too small. Show the exact working, as on slide 8.

Q5

> k-hop path, each link with BER p_b, packet of L bytes. What is the probability the packet arrives correctly?

1. The packet has 8L bits.
2. On one link, all the bits are good with probability $(1 - p_b)^{8L}$.
3. The packet must survive all k links, so multiply k times.

$P(\text{correct}) = (1 - p_b)^{8Lk}$

> So $P(\text{error}) = 1 - (1 - p_b)^{8Lk} \approx 8Lk \times p_b$ when that is small.

Q6

> 5-hop path, BER 10^{-5}, packet of 1,200 bytes. What is the probability the packet is in error over the whole path?

1. Bits: $1{,}200 \times 8 = 9{,}600$. Over 5 hops: $9{,}600 \times 5 = 48{,}000$.
2. Approximation: $48{,}000 \times 10^{-5} = 0.48$. That's too big, so use the exact formula.
3. Exact: $1 - (1 - 10^{-5})^{48{,}000} = 1 - 0.6188 = 0.381$.

≈ 0.38 (38%)

Q7

> Same path as Q6. What is the maximum packet size in bytes for an error probability below 1 in a hundred?

1. Condition: $1 - (1 - 10^{-5})^{8L \times 5} < 0.01$.
2. Total bits: $40L < \ln(0.99) \div \ln(1 - 10^{-5}) = 1{,}005$ bits. (The approximation gives $0.01 \div 10^{-5} = 1{,}000$ bits.)
3. $L < 1{,}005 \div 40 = 25.1$, rounded down to 25 B.
4. Check: 25 B gives 0.995% ✓ and 26 B gives 1.03% ✗.

25 bytes

Q8

> 8 intermediate nodes, all links with the same BER, packet of 1,250 B. What is the highest BER so that 99.9% of packets arrive correctly?

1. Hops = 8 + 1 = 9.
2. Bits: $1{,}250 \times 8 = 10{,}000$. Over 9 hops: 90,000.
3. Condition: $(1 - p_b)^{90{,}000} \geq 0.999$.
4. Solve: $p_b \leq 1 - 0.999^{1/90{,}000} = 1.11 \times 10^{-8}$. (The approximation gives $0.001 \div 90{,}000 = 1.11 \times 10^{-8}$.)

$p_b \approx 1.1 \times 10^{-8}$